Question. A human resources research firm reports that 22% of employees at large companies plan to change jobs within the next year. An HR consultant randomly selects 500 employees at large companies. Find the probability that the sample proportion of employees who plan to change jobs is less than 20%. Show all steps using the SPDC structure.

Detailed prompts

S – State:

1. Define the population (true) parameter in context.
2. Define the sample parameter in context.
3. What is 500? What does it represent?
4. What is 22%? What does it represent?
5. Write the problem (the probability you are asked to find using $\hat{p}$).

P – Plan:

1. Check randomness: Was the sample random? Why does randomness matter?
2. Check independence (10% condition): How many employees at large companies must there be in the population? Is $n < 10\%$ of N?
3. Check normality (large counts condition): Is the expected number of successes at least 10? Is the expected number of failures at least 10?
4. Compute the mean of the sampling distribution of $\hat{p}$.
5. Compute the standard deviation of the sampling distribution of $\hat{p}$, and write the Normal model.

D – Do:

1. Write the sampling distribution of the sample statistic.
2. Write the probability statement in terms of $\hat{p}$.
3. Write the probability statement in terms of Z formula.
4. Convert the boundary to a z-score.

5. Use the standard Normal table and write the value of the probability.

C – Conclude:

1. Start your conclusion with "There is about a ___ probability that…"
2. What is the sample size, and who or what is in the sample?
3. What statistic does the sample produce?
4. What value is in the question?
5. What does this proportion describe in context?